

PRESENTS

THE PETRIDISH

Culturing the future of Biotechnology by **BioSoc** | DTU

HIGHLIGHTS

History of Biotechnology

From ancient fermentation to synthetic genomes, biotechnology reflects the evolving dialogue between humans and life itself.

The Pivot Point

The critical turning moment where a discovery or insight shifts direction, transforming understanding and unlocking new possibilities in biotechnology.

Scientists in Spotlight

Highlighting the pioneers whose discoveries are shaping the future of biotechnology.

Case Study

Real-world applications that show how biotechnology solves complex biological and medical challenges.

Essentials of Biotechnology:

Core concepts and fundamental principles that form the foundation of modern biotech.

Events in Spotlight

Key breakthroughs, innovations, and global developments driving the biotech field forward.

Member's Achievements

Milestones and accomplishments that demonstrate progress and impact in biotechnology.

Internship Opportunities

Opportunities to gain hands-on experience and explore careers in biotechnology.



Living Continuum

Welcome to The PetriDish, a place for intellectual people to nurture their ideas in biotechnology. It is more than just a newsletter; it is a carefully chosen ecosystem that turns the complexity of the living world into the language of innovation. It doesn't see biology as a fixed science, but as an interesting, changing field that will be important in the future.

Biotechnology is not a recent innovation; it is a continuum that extends from the inception of civilization to the intricacies of contemporary genomics. From the Sumerians harnessing fermentation in 6000 BC to the 21st-century synthesis of artificial genomes and mRNA vaccines, our relationship with biological processes has evolved from observation to active design. In this edition of The PetriDish, we explore this vast timeline, spotlighting the pioneers who are bridging the gap between biological systems and advanced technology. The PetriDish is built to help the next generation of ideas grow.

***The language of life is written in the sequence of DNA,
and our task is to learn how to read and write it.***

**~HAR GOBIND KHORANA
(BIOCHEMIST & NOBEL LAUREATE)**

HISTORY OF BIOTECHNOLOGY

From Fermentation to Gene Editing: Biotech Through the Ages

Biotechnology is not a recent invention — it is a continuum stretching from ancient civilisations to today's cutting-edge genome engineering. Humans have harnessed biological processes since as early as 6000 BC, long before the term itself was coined. Here we trace that remarkable journey and spotlight the brilliant pivots that changed the course of medicine and science forever.

The BioTech Timeline:

~6000 BC

Sumerians & Babylonians harness yeast fermentation to brew beer observation-driven, not science-driven.

~4000 BC

Egyptians use natural oils and resins for mummification accidentally preserving ancient pathogens for modern study.

1855

E. coli discovered — now a cornerstone of biotech research and production worldwide.

19th C.

Pasteur links microorganisms to fermentation and disease. Mendel unlocks the laws of inheritance through pea plant experiments.

1919

Karl Ereky coins the term "biotechnology" in Hungary describing the conversion of raw materials into useful products.

2008

J. Craig Venter synthesises an artificial bacterial genome and transplants it into a recipient cell.

2010

Doudna & Charpentier introduce CRISPR-Cas9, enabling precise DNA editing in any organism.

2020

FDA approves the first mRNA COVID-19 vaccines — a landmark in modern medicine and rapid biotech response.

HISTORY OF BIOTECHNOLOGY

1

Ancient Biotech:

Humans have used the concepts of biotechnology since the dawn of civilization. Early humans unknowingly used biological processes in everyday life for multiple purposes, such as:

1. **Around 6000 BC**, the Sumerians and Babylonians harnessed biotechnology and used it in the fermentation of yeast to brew beer. Their practice was based on observations and traditions rather than the science of it.
2. **In 4000 BC**, the Egyptians used natural preservatives, such as oils and resins, to preserve bodies; this process was known as mumification. Interestingly, this process also led to the preservation of ancient pathogens. These pathogens provided a unique opportunity for scientists to examine and understand the diseases of the past with modern biotechnological tools.

2

Classical Biotechnology:

The term “biotechnology” was coined by Karl Ereky in 1919 in Hungary. The term was used to describe a technology based on converting raw materials into a more useful product.

In the 19th century, there was a breakthrough in scientific discoveries. These include:

1. Louis Pasteur demonstrated that microorganisms drive fermentation and disease.
2. Gregor Mendel discovered the basic principles of inheritance through pea plant experiments.
3. In **1855**, an extremely important bacterium was discovered, the Escherichia Coli bacterium. It has now become a major research, development, and production tool for biotechnology.

3

Modern Biotechnology:

The dawn of the 21st century brought forth a new and transformative era in biotechnology. Genetic engineering was finally introduced and it became the basis of new biotechnology.

1. In **2008**, J. Craig Venter and his team accomplished an incredible feat - they synthesised an artificial bacterial genome and transplanted it into a recipient cell.
2. In **2010**, Jennifer Doudna and Emmanuelle Charpentier brought about a game-changer - the CRISPR Cas9 system. It revolutionized genome editing by enabling precise alterations in DNA sequences.
3. In **2020**, the FDA approved the first COVID-19 vaccines developed using innovative messenger RNA (mRNA).

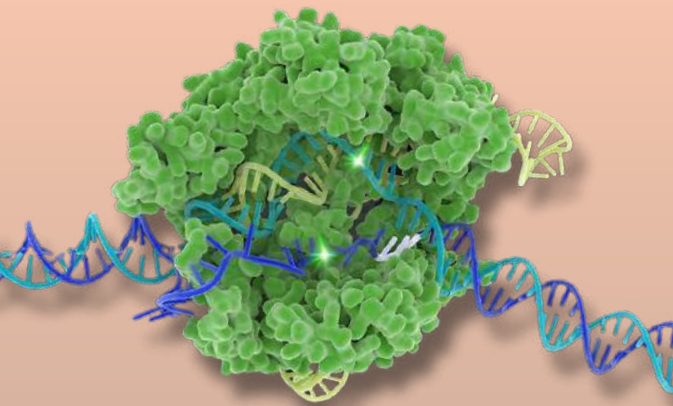
THE PIVOT POINTS

AZT: Failed cancer drug that fought HIV

Original goal: AZT, or azidothymidine, was originally developed in the 1960s by a U.S. researcher as a way to thwart cancer; the compound was supposed to insert itself into the DNA of a cancer cell and mess with its ability to replicate and produce more tumor cells. But it didn't work when it was tested in mice and was put aside.

Pivot: During the 1980s HIV/AIDS crisis, researchers desperately screened existing compounds for antiviral activity. They "dusted off" AZT and found it was a potent inhibitor of the virus and a treatment for HIV.

Impact: AZT was heralded as a "breakthrough" and "the light at the end of the tunnel", and pushed the FDA to approve the first AIDS medication on March 19, 1987, in a record 20 months.



From Bacteria to Bio-Scissors

Original goal: In 2006, this research began as a curiosity-driven exploration into a mysterious genetic mechanism found within certain bacteria called CRISPR. At the time, the team was simply trying to understand how microbes store genetic memories of past viral infections to protect themselves from future attacks.

Pivot: In 2012, the breakthrough occurred when they discovered that by providing the Cas9 protein, used by bacteria to fight viruses, with a custom-engineered RNA guide, they could program it to seek out and cut any specific DNA sequence in any organism.

Impact: This pivot transformed a niche bacterial immune process into a universal gene-editing toolkit, which is now being used to revolutionize medicine, agriculture, and biotechnology.

Penicillin: The first Antibiotic

Original goal: In **September 1928**, bacteriologist Alexander Fleming was conducting research at St. Mary's Hospital in London. He was studying Staphylococci, a common group of bacteria that causes various human infections. His objective was to observe the growth patterns of these bacteria and investigate potential ways to control or treat the illnesses they produced.

Pivot: Before leaving for a two-week vacation, Fleming left several culture plates on his laboratory bench. Upon his return, he found that one of the plates had been accidentally contaminated with a blue-green mold called *Penicillium notatum*. Instead of immediately cleaning the ruined sample, he noticed a clear zone around the mold where the bacteria had been destroyed. He correctly hypothesized that the mold was secreting a substance—which he named Penicillin, that was capable of killing the pathogens.

Impact: This accidental discovery introduced the world's first antibiotic. It fundamentally shifted medicine from a stage where simple infections could be fatal to one where bacterial diseases were reliably curable. Later refined for mass production by Howard Florey and Ernst Chain, Penicillin saved millions of lives during World War II and established the foundation for the modern pharmaceutical industry's work in life-saving antimicrobial therapies.

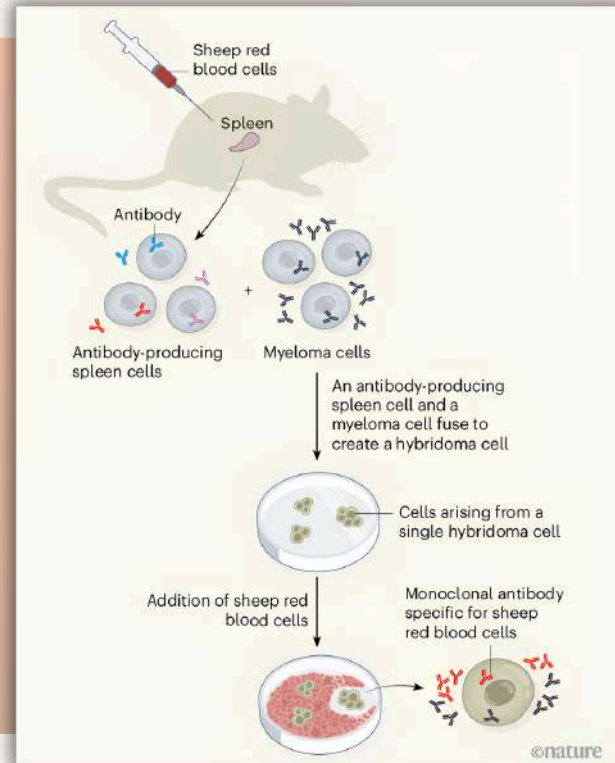


Monoclonal Antibodies

Original goal: Georges Köhler and César Milstein were trying to understand how antibody diversity is generated by the immune system. They were frustrated because B cells (which make antibodies) die quickly in lab cultures.

Pivot: They overcame the difficulty of growing antibody-producing B cells in culture by fusing spleen cells derived from mice immunized with a specific antigen to immortalize myeloma cells, a technique that enabled production of antigen-specific antibodies in culture indefinitely. They expected to just study the resulting hybrid cell's genetics.

Impact: They realized they had created a factory for monoclonal antibodies. This pivot created a multi-billion dollar industry of targeted therapies for cancer and autoimmune diseases (like Humira or Herceptin).

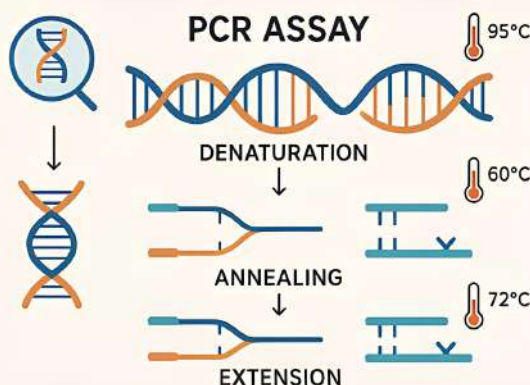


Polymerase Chain Reaction

Original goal: In 1983, Kary Mullis was working at Cetus Corporation, trying to develop a way to use oligonucleotides to detect point mutations in DNA, a slow and frustrating process.

Pivot: During a night drive in 1983, he had a "lightbulb moment." He realized that by using two primers and a polymerase in a repeating cycle, he could exponentially amplify any segment of DNA rather than just detecting it.

Impact: He pivoted from diagnostic detection to PCR (Polymerase Chain Reaction). This technology automated the process of DNA replication, enabling scientists to produce millions of exact genetic copies in a few hours and removing the requirement for weeks of manual laboratory labor.



SCIENTISTS *in SPOTLIGHT*



Martina Pukhrambam

“Pioneering Yak Genomics to Advance Climate-Resilient Livestock and Himalayan Biotechnology”

Martina Pukhrambam is an Indian scientist who stands out as a remarkable scientific mind in the field of animal genetics and biotechnology. Her contribution in decoding the Indian yak genome is a revolutionary breakthrough in modern science. As a key researcher in ICAR- National Research Centre on Yak, Dr. Pukhrambam played a prominent role in successfully achieving the first- ever high- quality, chromosome- level genome sequencing of the Indian yak. This is a milestone in Indian scientific achievement, as it explains how yaks survive in extreme temperatures, resists diseases, and stay efficient in harsh conditions. Through this genome sequence we can change and improve livestock breeding, find out different conservation strategies, and sustain agriculture in the Himalayan region. Dr. Pukhrambam’s accomplishment is recognized as a big leap forward for Indian biotechnology.



Bhanendra Sahu

“Pioneering 4-D Smart Polymers for Adaptive and Minimally Invasive Biomedical Innovations”

Bhanendra Sahu is an Indian innovator in the field of biomedical engineering and advanced materials. His work on 4-D smart polymers demonstrates a remarkable leap in modern science. 4-D smart polymer is a next- generation material that not only exists in three dimensions but also changes its properties or shape over time in response to external stimuli. This material is extremely useful for biomedical applications as it holds immense promise for minimally invasive medical treatments, in which implants or devices made up of such polymers can expand, reshape or activate precisely when needed. His achievement showcases a brilliant example of futuristic science, carrying the potential to revolutionize biomedical treatments and save countless lives.



John Donoghue

“Pioneering brain–computer interfaces that transform neural intent into action, redefining independence for paralysis.”

John Philip Donoghue is an American neuroscientist who is a renowned figure whose work has connected the human brain and technology. He is well known for developing “Brain Gate”, a brain-computer interface (BCI) system that has fascinated the scientific world. Brain Gate helps restore movement for people with paralysis and with prosthetics. It involves the use of a small, implanted sensor in the motor cortex to capture neural signals, which allows the user to control computers or robotic arms through thought alone. The impact of Brain Gate is monumental, and it lays the foundation for future innovations in neuroprosthetics and human-machine integration. Brain Gate not only demonstrates an achievement in neuroscience and biomedical engineering, but it also instills hope for many who face physical limitations.



Dr. Anna Trigos

“Decoding cancer’s networks to restore cellular harmony and guide precise targeted therapies.”

Anna Trigos is an Australian–Argentine computational biologist who uses large-scale “omics” data and network-based models to study cancer as an evolving ecosystem. By integrating genomic, transcriptomic, and other molecular data across thousands of tumor samples, she has shown how regulatory networks become rewired in cancer, disrupting communication between multicellularity-maintaining pathways and “selfish” cellular programs. Her work reveals that many cancer-linked mutations target systems that normally control cell behavior, offering new insights for tumour-driver networks and targeted therapies, while advancing bioinformatics approaches for clinically relevant cancer research.

CASE STUDIES

1 The "Sushi-Roll" Biohybrid Hand



Biometric Humanoid Hand Driven by Large-Scale Muscle Tissue Actuators (MuMuTAs) by Professor Shoji Takeuchi (University of Tokyo).

Summary: The study solved the necrosis problem in lab-grown tissue by rolling thin sheets of human muscle cells into cylindrical "sushi-roll" structures, allowing nutrients to reach the core. This enabled the creation of larger, functional muscle bundles capable of powering a humanoid hand—showing that living tissue can act as a mechanical driver in robots.

Process: Takeuchi's team developed this using a novel "sushi-roll" muscle engineering technique.

MuMuTAs (Multiple Muscle Tissue Actuators): Instead of growing one thick muscle, researchers grew thin, flat sheets of human muscle fibers in a culture medium.

The Rolling Technique: These sheets were rolled into cylindrical bundles, similar to how ingredients are wrapped in a sushi roll. This structure maximized contractile force while ensuring every cell remained close enough to the culture medium to receive oxygen and nutrients.

Mechanical Integration: These "muscle tubes" were mounted on a 3D-printed plastic forearm and connected to multi-jointed fingers via a cable-driven, tendon-like mechanism.



Achievements and Results

The biohybrid hand showed advanced, life-like performance:

- **Strong output:** Each MuMuTA bundle generated about 8 mN of force, enabling precise finger movement
- **Complex actions:** It performed gestures like "scissors" and handled small objects (e.g., a pipette tip)
- **Biological behavior:** The muscles showed fatigue after ~10 minutes but fully recovered after rest, similar to real human muscle

2 The "Third Layer" of Jaw Muscle

Identification of a Deep, Third Layer of the Human Masseter Muscle

Head Scientist & Co-Lead

Dr. Szilvia Mezey (University of Basel, Department of Biomedicine) & Professor Jens Christoph Türp (University Center for Dental Medicine Basel)

Before the Discovery: The "Two-Layer" Dogma

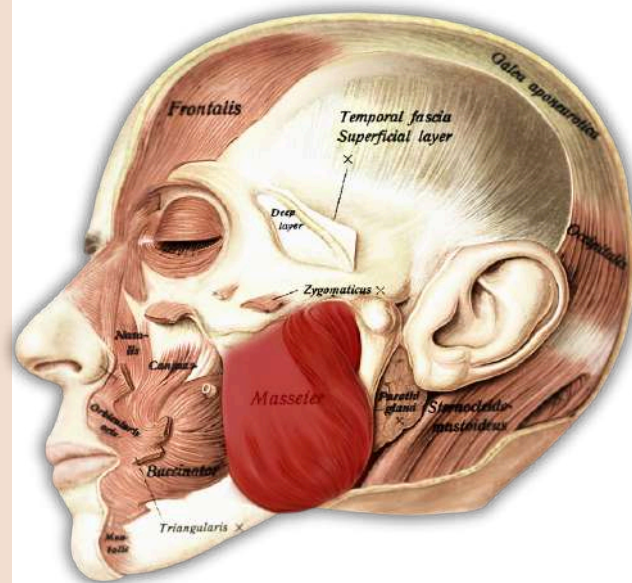
For over 150 years, the masseter muscle was described as having two layers: superficial and deep. This model guided dental and jaw treatments, but it couldn't fully explain certain movements like slight backward (retraction) motion of the jaw.



Dr. Szilvia Mezey and her team carried out a detailed, multi-step study to re-examine the structure of the masseter muscle:

- **High-resolution MRI:** They used advanced imaging on living subjects to observe how muscle fibers are arranged and function during active jaw movement.
- **Detailed dissection:** The team dissected 12 formaldehyde-preserved human cadaver heads, with special focus on the coronoid process of the mandible to study deeper muscle attachments.
- **Histological validation:** Tissue cross-sections were analyzed to confirm that the fiber arrangement and nerve supply were distinct from the known superficial and deep layers.

This combined approach provided strong evidence that the masseter is more complex than the traditional two-layer model.



What They Achieved: Real-Time Data & Discovery

In December 2021 (with validation through 2024), Dr. Szilvia Mezey's team identified a third, deeper layer of the masseter called the pars coronidea.

- **Distinct anatomy:** originates from the zygomatic bone and inserts into the coronoid process of the mandible
- **Unique function:** unlike the other layers that close the jaw, it pulls the jaw backward (retraction) and stabilizes the TMJ
- **Findings:** present in 100% of cadaver samples; smaller than other layers but crucial for fine control of jaw movement

3

The Identification of "Pliancy Genes" in Human Tissue (2026)



Epigenetic Reprogramming and the Activation of Latent Pliancy Genes for Retinal and Neural Recovery

Head Scientist

Dr. Michael A. Dyer (Chair of the Department of Developmental Neurobiology at St. Jude Children's Research Hospital)

The Status Quo: Before the Discovery

Until early 2026, the retina and brain were believed to be non-regenerative, so damage from injury or disease was considered permanent.

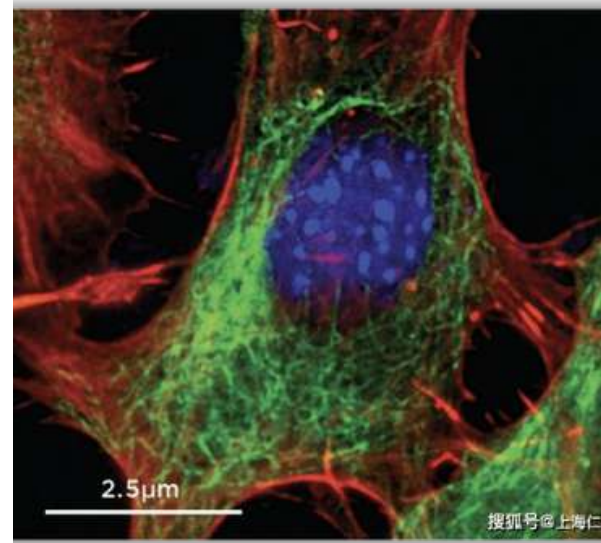
The Discovery: Finding the "Hidden" Software

In early 2026, Dr. Michael Dyer and his team at St. Jude Children's Research Hospital identified "pliancy genes"—ancient genetic programs in humans that are normally switched off by epigenetic controls.

How They Did It?

- **Cell comparison:** Used single-cell sequencing to compare zebrafish (regenerative) and human retinal cells
- **Mapping the gap:** Found zebrafish cells revert to stem cells, while human cells form scar tissue (gliosis)
- **Gene identification:** Discovered key genes (Ascl1, Lin28) exist in humans but are blocked by protein complexes

The Intervention: Using CRISPR-based epigenetic editing, the team removed molecular "padlocks" to reactivate dormant regenerative genes without altering the DNA.



What They Achieved

- **Regeneration:** Treated retinal organoids showed ~45% recovery of photoreceptors in 60 days (vs 0% in control)
- **Cell conversion:** ~15% of support cells turned into functional neurons
- **Integration:** New neurons actively fired signals with ~72% efficiency of healthy tissue

The Breakthrough

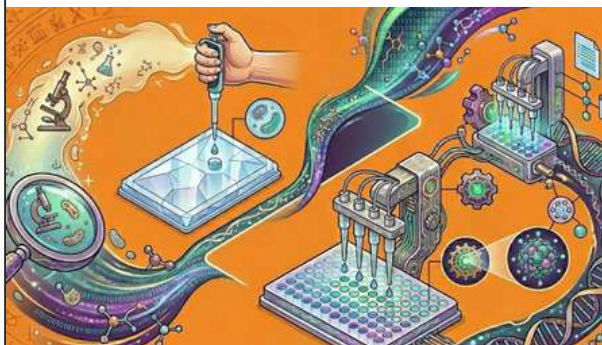
The discovery of pliancy genes shifts medicine from "replacement" to "reawakening"—unlocking the body's own repair systems.

For bio-humanoid tech, this means tissues could self-repair, adapt, and integrate, enabling more lifelike and responsive prosthetics or robotic systems

ESSENTIALS OF BIOTECHNOLOGY

1

Wet-Lab Mastery: The "Precision" Revolution



The Landscape: Traditional "wet lab" work is evolving into high-throughput biology. It is no longer enough to just run a gel; the modern researcher must understand how to interface with automation and microfluidics.

Current Research Spotlight: A major area of focus is Single-Cell Multi-omics. Researchers are no longer looking at tissues as a whole but are using tools like 10x Genomics platforms to sequence the RNA and chromatin accessibility of individual cells simultaneously. This is critical in Oncology research, where scientists are trying to understand "tumor heterogeneity"—why some cells in a single tumor resist chemotherapy while others die. By mastering technologies such as Next-Generation Sequencing or organ-on-a-chip cultivation, you move from being a lab assistant to a specialised Biotech-engineer.

Why Invest Time: Companies like Illumina and Thermo Fisher are shifting toward "Sample-to-Insight" workflows. If you master advanced techniques like Library Preparation for Next-Gen Bio-Engineer.

How to Start: Start by mastering the "Golden Gate" or "Gibson Assembly" methods for cloning. These are the industry standards for building complex DNA circuits. Look into the iGEM (International Genetically Engineered Machine) archives to see how students are using these tools to build microbes that can sense toxins or produce medicines.

2

Bioinformatics & Data Science: The "Dry Lab" Engine

The Landscape: We are producing biological data faster than we can analyze it. The bottleneck in biotech is no longer the experiment; it is the interpretation.

Current Research Spotlight: The integration of Large Language Models (LLMs) like Evolutionary Scale Modeling (ESM-2). Much like ChatGPT predicts the next word in a sentence, researchers at Meta AI and DeepMind are using AI to predict the "next" amino acid in a protein sequence to create entirely synthetic enzymes that don't exist in nature. This "Generative Biology" is being used to design enzymes that can break down plastics in hours instead of centuries.

Why Invest Time: Statistics and Python are the new "pipettes." If you can write a script to scrape the NCBI database or perform a Genome-Wide Association Study (GWAS), you become invaluable to clinical research organizations (CROs).. Wet-Lab Mastery: The "Precision" Revolution.

How to Start: Don't just learn "Python"; learn "Tidyverse" in R or "Scanpy" in Python. These are specifically designed for biological data. Participate in Kaggle competitions or solve problems on Rosalind.info to build a portfolio that proves you can handle "noisy" biological data.



3

Structural Biology & Protein Engineering: Designing the "Lock and Key"



The Landscape: For decades, solving a protein structure took years of X-ray crystallography. Now, it takes minutes.

Current Research Spotlight: The move from AlphaFold2 to AlphaFold3 and RoseTTAFold. These AI systems can now predict the structures of not just proteins, but also DNA, RNA, and "ligands" (drug molecules). Recent research published in Nature shows scientists using these tools to design "De Novo" binders—protein "cages" that can trap viruses like Influenza or COVID-19 before they enter human cells.

Why Invest Time: This is the heart of the Pharmaceutical industry. If you understand molecular docking and protein-protein interactions (PPI), you can work in Computer-Aided Drug Design (CADD). This is where the highest-paying roles in biotech currently reside.

How to Start: Download PyMOL (there is a free version for students) and start visualizing proteins from the PDB (Protein Data Bank). Learn how to perform "Molecular Dynamics" (MD) simulations using tools like GROMACS. Seeing how a protein "wiggles" in water is the first step to understanding how it functions in the body.

Computational Genomics & System Phylogenetics: Decoding the Tree of Life

The Landscape: Bioinformatics has moved beyond simple sequence matching. We are now in the era of Pan-genomics and Metagenomics, where we don't just sequence one organism, but entire ecosystems—from the human gut microbiome to the soil of the Amazon rainforest. The challenge is no longer just "reading" DNA, but reconstructing the complex evolutionary history and functional networks of thousands of species simultaneously.

Current Research Spotlight: A groundbreaking area is Metagenomic Pathogen Surveillance. Researchers are currently using "Environmental DNA" (eDNA) from city wastewater to detect viral outbreaks—like Polio or new SARS-CoV-2 variants—weeks before patients even show up in hospitals. By using Phylodynamic modeling, scientists can trace the exact mutation path of a virus in real-time. Another massive leap is Ancient DNA (aDNA) Genomics. Recent studies in *Science* and *Nature* have successfully sequenced DNA from million-year-old mammoth teeth and 2,000-year-old human remains to understand how ancient pandemics shaped our modern immune systems.

Why invest time: As the world prepares for "Disease X" (the next potential pandemic), bioinformatics specialists who can handle comparative genomics and phylogenetic tree construction are in high demand by global health organizations like the WHO and the CDC. Furthermore, in the agricultural sector, pan-genomics is being used to identify "climate-resilient genes" in wild relatives of crops like wheat and rice, which is a multi-billion dollar industry aimed at ensuring global food security.

How to Start: Start by learning the Linux Command Line (Bash); almost all high-level genomic tools run on Linux. Familiarize yourself with tools like MEGA11 for phylogenetics and QIIME 2 for microbiome analysis. Explore the Ensembl Genome Browser to see how different species' genomes are mapped and compared. If you can build a "Maximum Likelihood" tree and explain what the "Bootstrap value" means, you have a solid foot in the door of evolutionary and clinical research.



EVENTS IN SPOTLIGHT

1 *BioInsight Forum*

~ (February 27th, 2026)

BioSoc-DTU initiated Invictus Fest with the BioInsight Forum, a thought-provoking panel discussion centered on "AI in Drug Discovery." The event brought together experts from academia, healthcare, and industry to explore how artificial intelligence is transforming modern medicine. Through discussions on lead optimization, clinical applications, and ethical challenges, the forum bridged the gap between theoretical knowledge and real-world biotech innovation, inspiring students to explore interdisciplinary careers.



2 *Lab Rats Quiz 4.0*

~ (February 28th, 2026)



BioSoc-DTU organized Lab Rats Quiz 4.0 under Invictus, bringing together biotechnology enthusiasts for an engaging and competitive quiz experience. The event tested participants on core biotech concepts through multiple rounds including elimination, buzzer, image-based, and rapid-fire formats. With strong participation from various institutions, the quiz fostered scientific thinking, teamwork, and quick problem-solving, making it a dynamic platform for learning beyond textbooks.

3 *Ecopreneurs Case Competition*

~ (March 1st, 2026)

BioSoc-DTU organized the Ecopreneurs Case Competition, encouraging students to tackle sustainability-focused challenges through innovative and research-driven solutions. Participants analyzed real-world problems inspired by the pharmaceutical industry and presented strategic approaches involving biomarker validation, clinical trials, and precision medicine. The competition promoted analytical thinking, entrepreneurship, and practical problem-solving while providing valuable exposure to expert evaluation and industry insights.



4 *BioHackathon 2.0*

~ (March 1st, 2026)



BioSoc-DTU conducted BioHackathon 2.0, a highly technical and application-based event featuring tracks in AI/ML and RNA-Seq data analysis. Participants worked on real biological datasets, building predictive models and solving sequence-based problems using advanced bioinformatics tools like BLAST, STAR, and DESeq2. The hackathon emphasized computational skills, data interpretation, and innovation, offering hands-on experience in integrating machine learning with biological research.

5 *Drug Discovery Using AI and In Silico Methods Workshop*

~ (April 8th, 2026)

BioSoc-DTU expanded learning horizons with a hands-on workshop on drug discovery using AI and in silico tools. From molecular docking to AI-driven target identification, participants explored the practical side of computational biology. Using tools like PyMOL and PyRx, we performed real-time docking and visualized protein-ligand interactions, gaining insights into modern drug development workflows. The session also introduced databases like GEO for gene expression analysis, showing how biotechnology and artificial intelligence together are transforming faster, cost-effective drug discovery.





THE GROWTH CURVE: SPOTLIGHT ON EXCELLENCE

Rishabh Mohan Sinha (3rd Yr B.Tech Biotech)

President | Harvard Conference

Rishabh Mohan Sinha has attended the **Harvard HPAIR Conference at Harvard University, Massachusetts, USA**, a prestigious global platform for academic and policy discussions. Actively participated in sessions on global healthcare, diplomacy, and advancements in biotechnology. Engaged with **international** delegates, gaining exposure to diverse perspectives and emerging global challenges in science and policy.



Hunar Aggarwal (3rd Yr B.Tech Biotech)

Vice President | Research Intern

Hunar Aggarwal completed a **research internship at Ganpat University** focusing on antimicrobial resistance in *Klebsiella pneumoniae*. Developed a machine learning pipeline for target identification and analysis. **Presented the research findings through a scientific poster at a conference, effectively communicating complex data.** Gained hands-on experience in computational biology and interdisciplinary research methodologies.

Samihan Sharma (3rd Yr B.Tech Biotech)

General Secretary | Publication, Incubation & Startup

Samihan Sharma developed "**NeuroSakhi**," an EEG-based machine learning framework for the early detection of ADHD. Designed and implemented the complete analytical pipeline, from data processing to model development. **Presented the concept as a startup at IIT Kanpur**, receiving positive feedback for innovation and applicability. **The work is currently under review for IEEE publication**, indicating strong research and entrepreneurial potential.



Mukund Gupta (2nd Yr B.Tech Biotech)

Junior Council | Paper Presentation

Mukund Gupta has successfully presented a poster as the primary author on the review paper "**Compound-Specific Utility of Liposomal Delivery Systems for Drugs and Nutraceuticals: A Critical Review of Pharmacokinetic Outcomes and Translational Trade-offs**" at the XIVth Annual International Conference of Indian Academy of Biomedical Sciences (**IABSCON-2026**) (**ARTH 2026**), held from 24–26 March 2026. The conference was jointly organized by **BITS Pilani, K K Birla Goa Campus and IIT Goa, India.**

THE GROWTH CURVE: SPOTLIGHT ON EXCELLENCE

Shivam Chaube (2nd Yr B.Tech Biotech)

Junior Council | Paper presentation

Shivam Chaube presented a research paper titled “**Programmable Mammalian Cells Using CRISPR-Based Logic Gates**” at the **XIVth Annual International Conference** of Indian Academy of Biomedical Sciences (IABSCON-2026) (ARTH 2026), held at **BITS Pilani, K K Birla Goa Campus**, in association with **IIT Goa**. The study focused on the design of synthetic CRISPR-based gene circuits for precise detection of cancer biomarkers.



Charu Agrawal (1st Yr M.Sc Biotechnology)

Member | Quiz Competition

Charu Agrawal has secured **1st position in a quiz competition** organized by the Department of Biotechnology, DTU, on 24 March 2026 under the World Tuberculosis Day awareness program. Demonstrated strong conceptual understanding and quick problem-solving abilities in biotechnology. Showcased excellent analytical thinking and awareness of current developments in the field during the competition.

Mansi Goyal (1st Yr M.Sc Biotechnology)

Member | GATE/Quiz Competition

Mansi Goyal has achieved an **All India Rank (AIR) of 927 in GATE XL** with **50.67 marks**, with a **GATE score of 590/1000**, reflecting a strong grasp of interdisciplinary scientific concepts. Also secured **1st position in a quiz competition** organized by the Department of Biotechnology, DTU, on 24 March 2026, under the World Tuberculosis Day awareness program. Demonstrated the ability to apply knowledge effectively in competitive settings.



Vansh Ralli (3rd Yr B.tech Biotech)

Member | GATE Rank

Vansh Ralli has secured **AIR 859 in GATE Biotechnology** with **36.33 marks** and a **GATE score of 498/1000**, demonstrating strong subject knowledge and competitive performance. Showcased consistent preparation and a deep understanding of core biotechnology concepts. This achievement highlights analytical proficiency and the ability to perform under high-pressure examination conditions.



Varun Gautam (1st Yr B.Tech Biotech)

Member | Review Paper

Varun Gautam assisted in the **preparation of a review paper** by conducting an extensive literature survey on liposome-based drug delivery systems in breast cancer. Reviewed 12+ high-impact research papers and supported systematic data compilation and analysis. Also contributed to structuring the manuscript and ensured accuracy in referencing. Additionally, facilitated coordination by identifying relevant institutional contacts and managing research-related communication.



BioBridge: Internship Opportunities



1. ICMR-NIRRH – Research Internship (Biotechnology / Life Sciences)

- **Portal to Apply:** Official ICMR-NIRRH website/internship section (portal opens via a Google-form-style application; check “Training / Internship” tab at <https://www.nirrh.res.in> and linked notices).
- **Duration:** ~2 months (June–July 2026, project-wise).
- **Eligibility:** B.Sc / M.Sc students in Life Sciences / Biotechnology; often advertised for 2026 with “opening soon” but many rounds open in April–May.
- **Apply by:** Slots announced periodically; most recent 2026 batch opens in June–July 2026 announcements; keep checking the site as the form reopens in rolling windows.
- **Focus:** Research-driven training in reproductive biology, virology, and related biotech-aligned areas within a premier ICMR institute.



2. AIIMS Biotechnology / Life Sciences Interns (Biotech-oriented)

- **Portal to Apply:** Apply via the official AIIMS “Medical / Biotechnology Intern” posting (often through institutional portals or external job boards; entry is via an online form that stays open on a rolling basis).
- **Duration:** 1–3 months (May–August 2026, flexible windows).
- **Eligibility:** Undergraduate/postgraduate students in Biotechnology / Life Sciences / allied disciplines.
- **Apply by:** Rolling basis; applications are currently open for 2026-style biotech-oriented internships on AIIMS-linked portals.
- **Focus:** Clinical-lab-aligned research exposure in biotechnology settings, with mentorship and possible LOR, though unpaid.



3. CSIR Labs (CCMB / IGIB) – Summer Research Internships

- **Portal to Apply:** Individual lab portals (e.g., CCMB, IMTECH, IGIB “Students / Internships” pages) or via the India Science & Technology internships portal (<https://www.indiascienceandtechnology.gov.in/listingpage/internships>).
- **Duration:** 2–6 months (summer 2026 windows; varies by lab).
- **Eligibility:** B.Sc / M.Sc / B.Tech students in Life Sciences / Biotechnology; often with stipend (₹5,000+ in some labs).
- **Apply by:** Applications open in batches; many CSIR labs reopen their forms in April–May 2026; check each lab’s “Internship / Training” page for live links.
- **Focus:** Lab-based research in molecular biology, genomics, structural biology, and related biotech areas under CSIR-lab faculty.



4. TCS iON Virtual Internship – Life Sciences / Biotechnology-aligned

- **Portal to Apply:** <https://www.tcsion.com> → “Virtual Internships” for Life Sciences / Biotechnology.
- **Duration:** 2–4 weeks (self-paced, runs anytime during 2026).
- **Eligibility:** Any undergraduate or postgraduate student in Life Sciences / Biotechnology / allied fields.
- **Apply by:** Rolling basis; form is open throughout 2026 for this virtual internship.
- **Focus:** Online, project-based training in biotech-aligned life-science topics, with free certificate; more skill-oriented than wet-lab research-focused.

EDITOR'S NOTE



SHASHWAT JHA

"Just like a real petri dish, this newsletter became a vessel for growth — proving that the best ideas happen when we bridge the gap between disciplines."

Some two years ago, in the quiet haze of the summer break after my first year, Petridish was just an idea something Apurva Di and I had scribbled down btw conversations. We were not trying to create just another document; we wanted to build a space. A space where biotechnology would not feel distant or intimidating, rather something you could explore through stories, curiosity and shared perspectives.

Coming from Electrical Engineering, stepping into biotechnology felt like my own little experiment. And in many ways, Petridish lived up to its name. Just like a real petri dish, this newsletter became a vessel for growth and proving that the best ideas happen when we bridge the gap between the disciplines.

As I bid adieu to this desk and hand over the helm, I do so with a lot of faith in the team taking over. What makes a petri dish fascinating is that it never stays the same the culture keeps evolving, adapting, becoming something new. And I am genuinely excited to now sit on the other side, reading what comes next, seeing the fresh perspectives and the new "strains" of thought that will shape its future.

It has truly been a pleasure.

Signing off,
Shashwat Jha
(Former Editor)

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